MODULE 6 · AI & ML Foundations

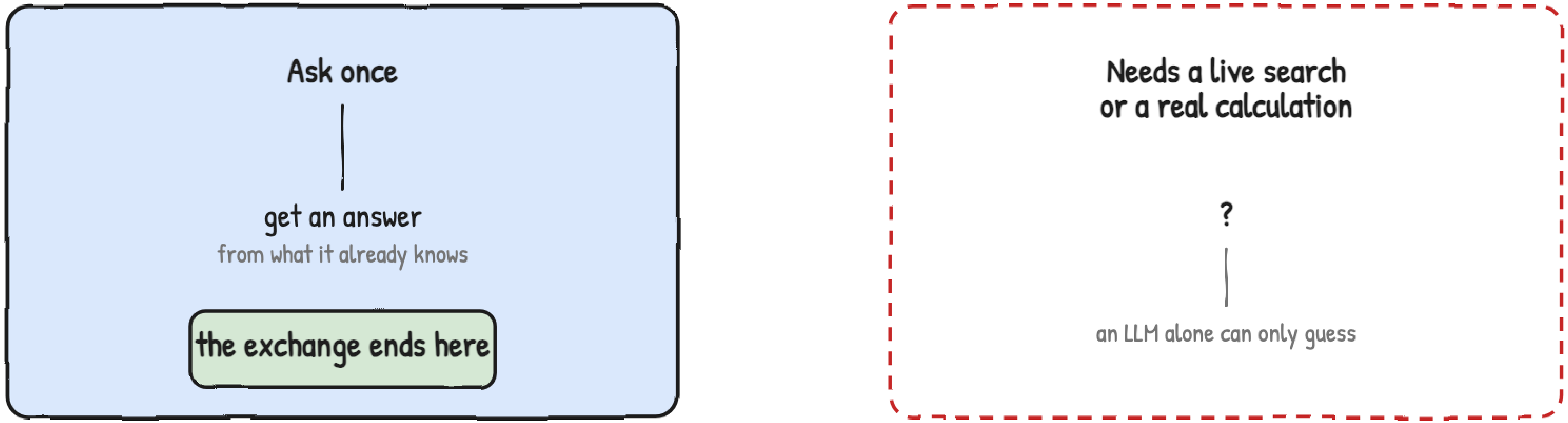
# Your First AI Agent — Tools, Memory & the Agent Loop

The agent Module 1 promised you'd build

**Gourav Shah** · School of DevOps & AI · Lesson + Lab + Quiz

What actually makes something an agent?

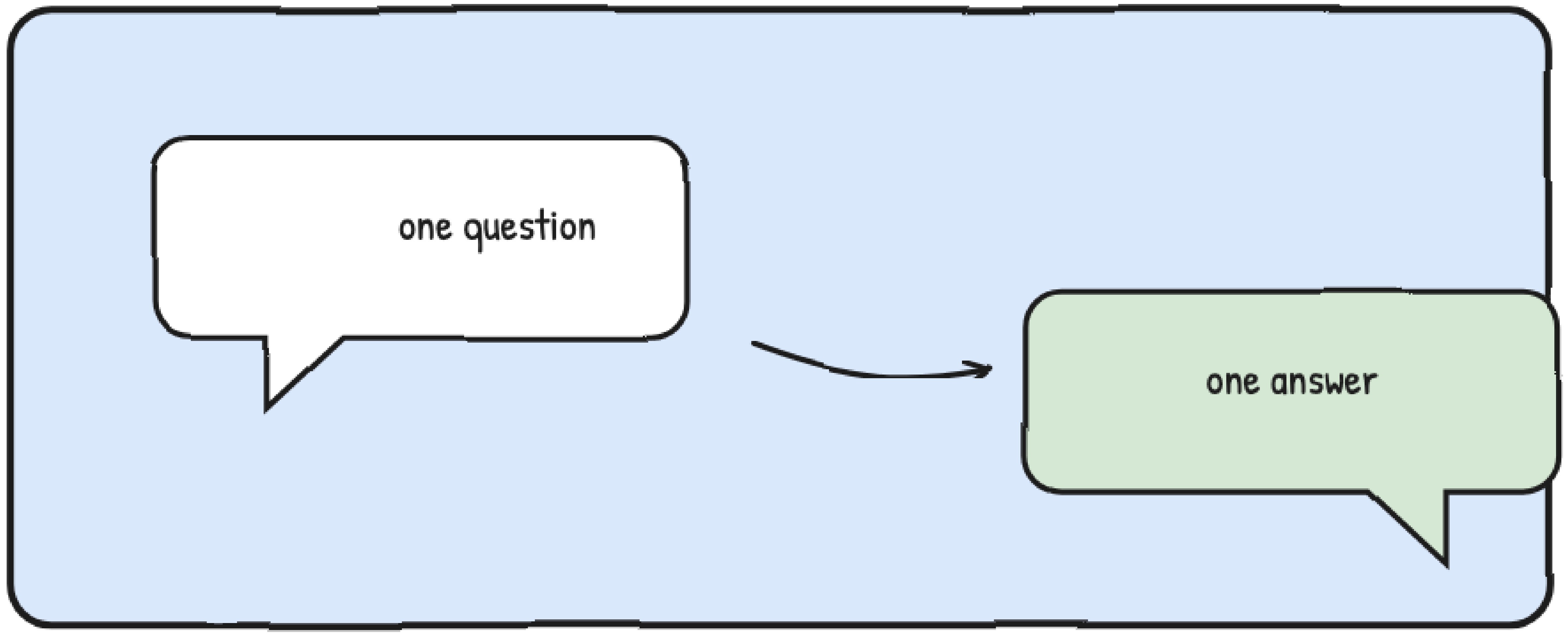
# A single call only gets you as far as what the model already knows




Module 5 answers once, and the exchange ends there

A model, a pipeline, or an agent?

# Ask a colleague once over chat — that's a model




the exchange ends here

One question, one answer, from what they already know

# A fixed checklist is a pipeline, no matter how many steps it has

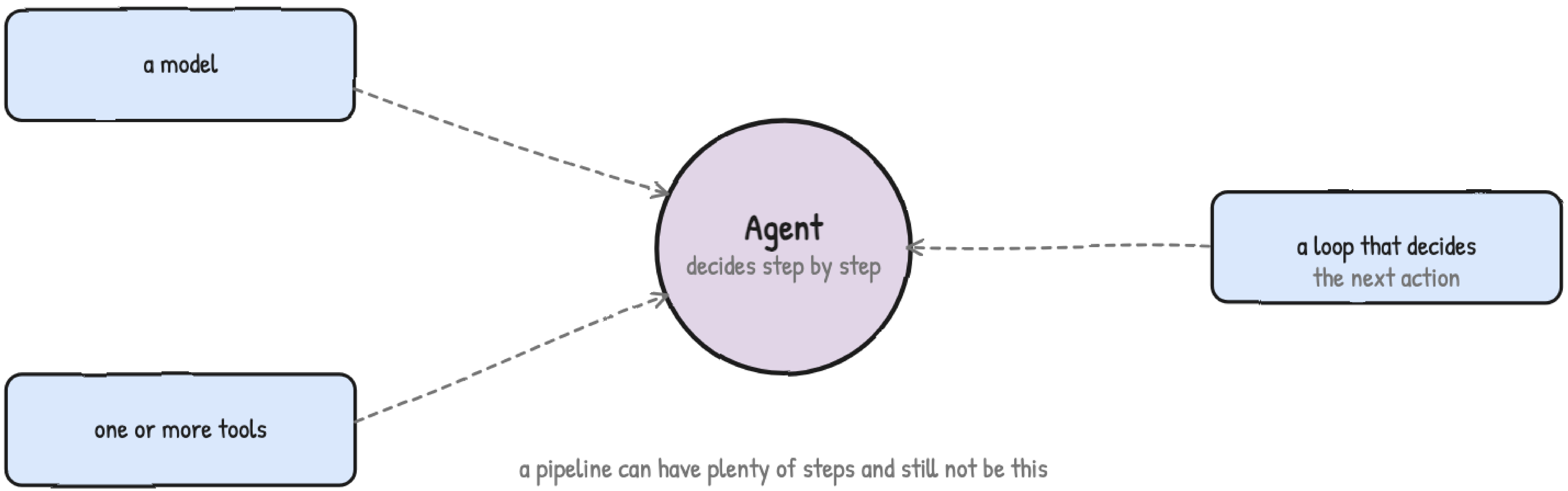

can look complicated and still not be an agent

Same order every time, decided in advance

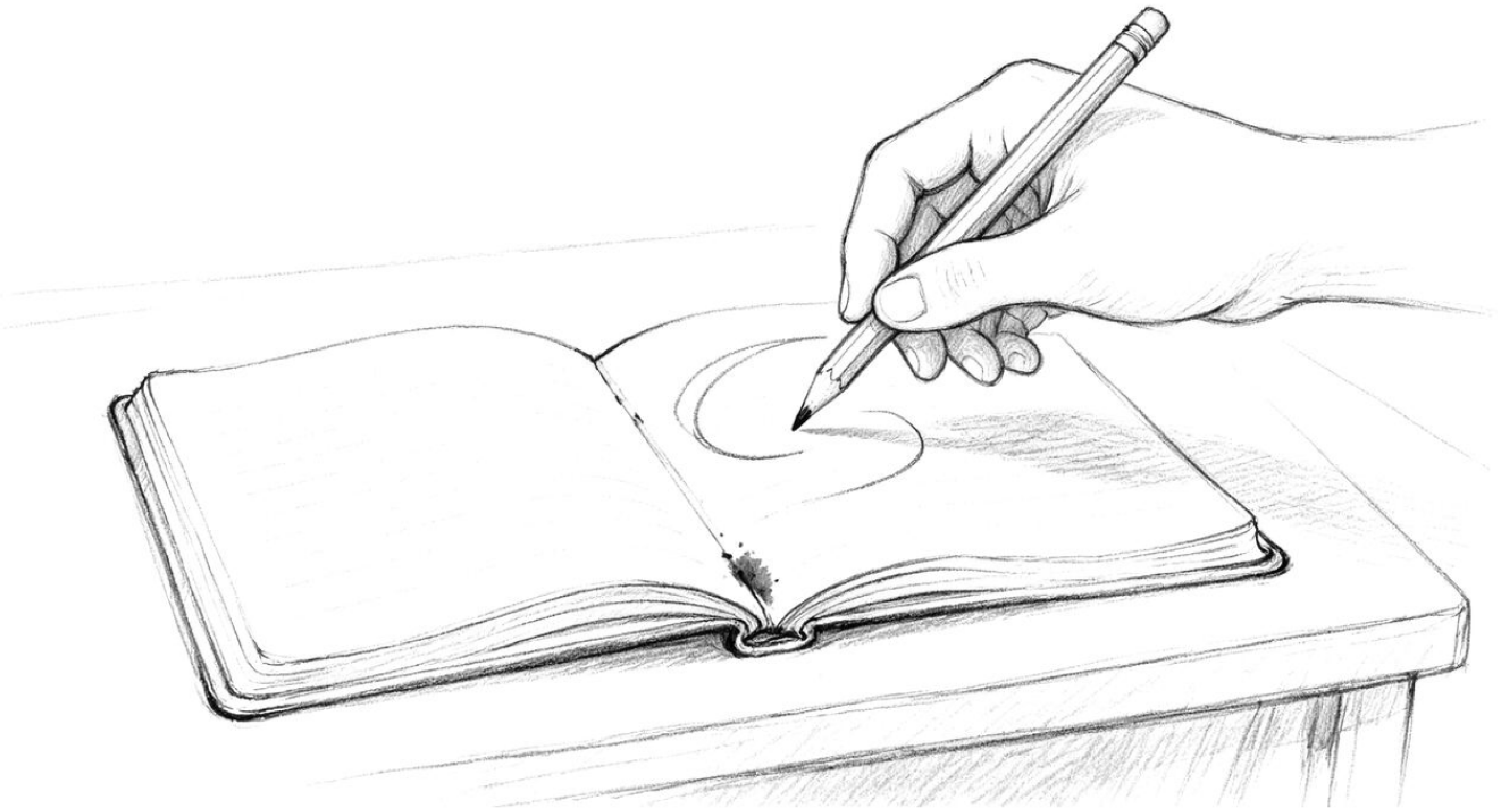
# Hand it to a capable teammate and let them figure out the steps

Decided step by step, not fixed ahead of time

# Wired to tools, running in a loop, deciding the next move

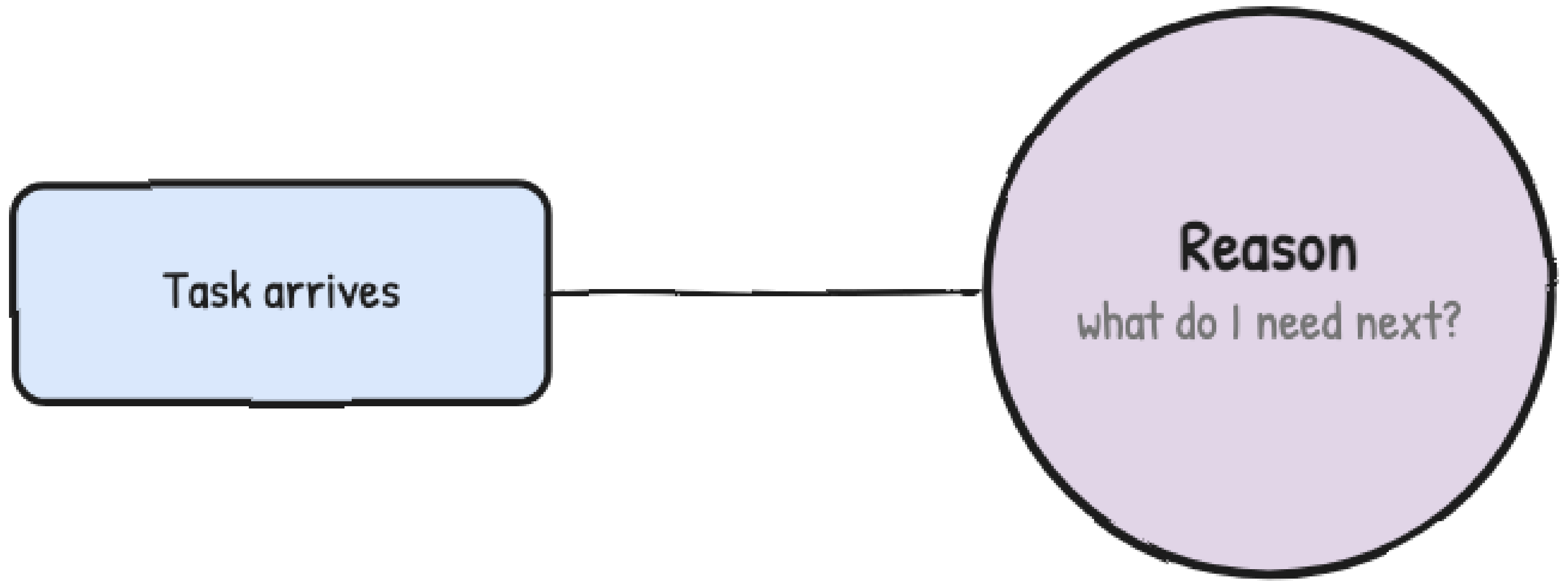



The three analogies, tied together into one definition



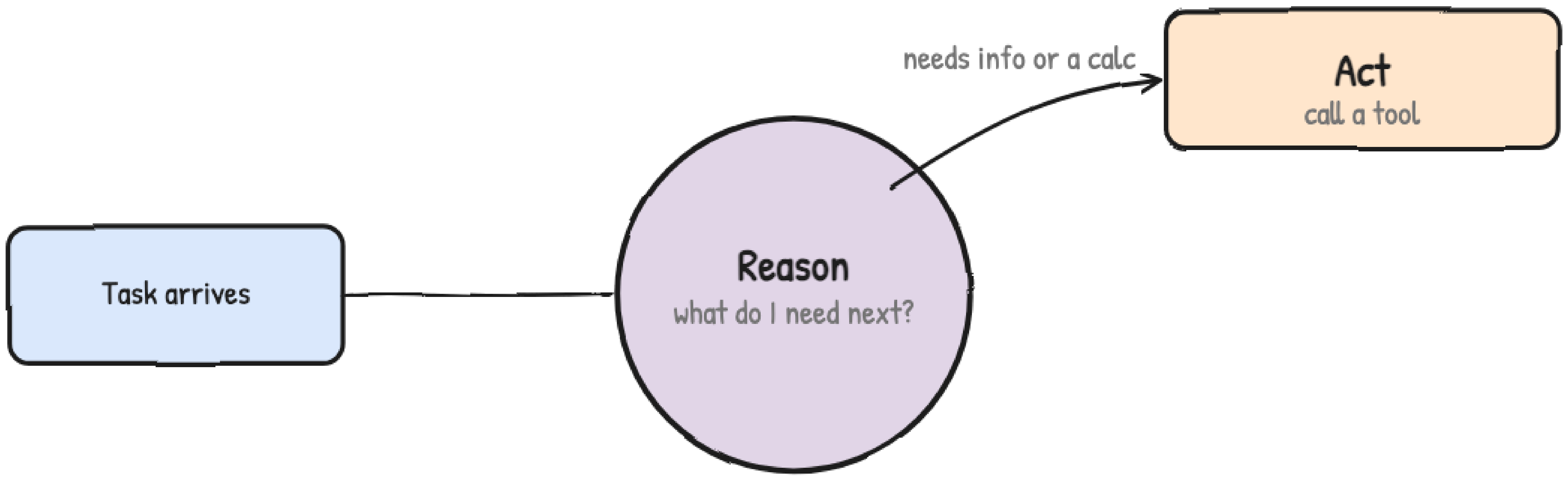
The agent loop: reason, act, observe, repeat

# One loop, run again and again, until the model has enough



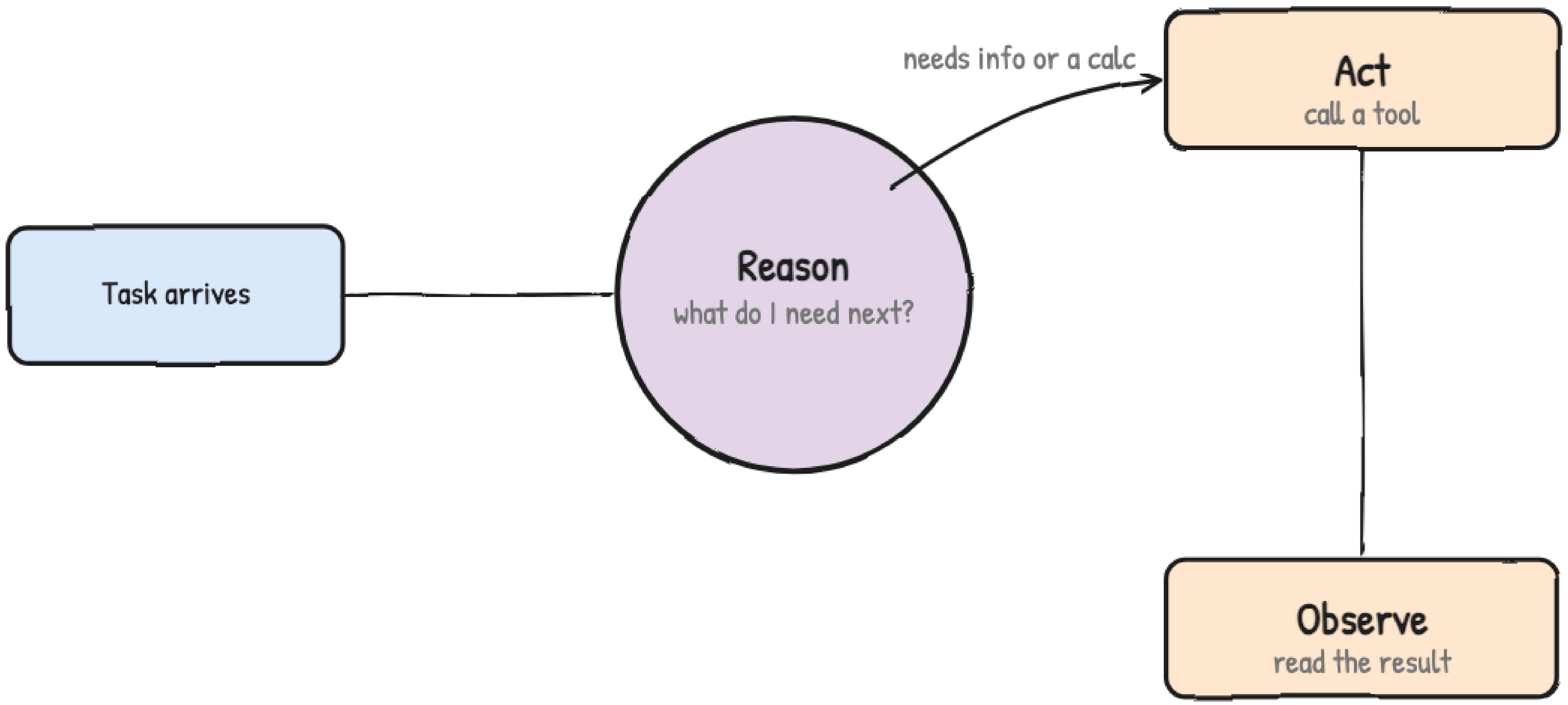

Every real agent is running this underneath

# One loop, run again and again, until the model has enough



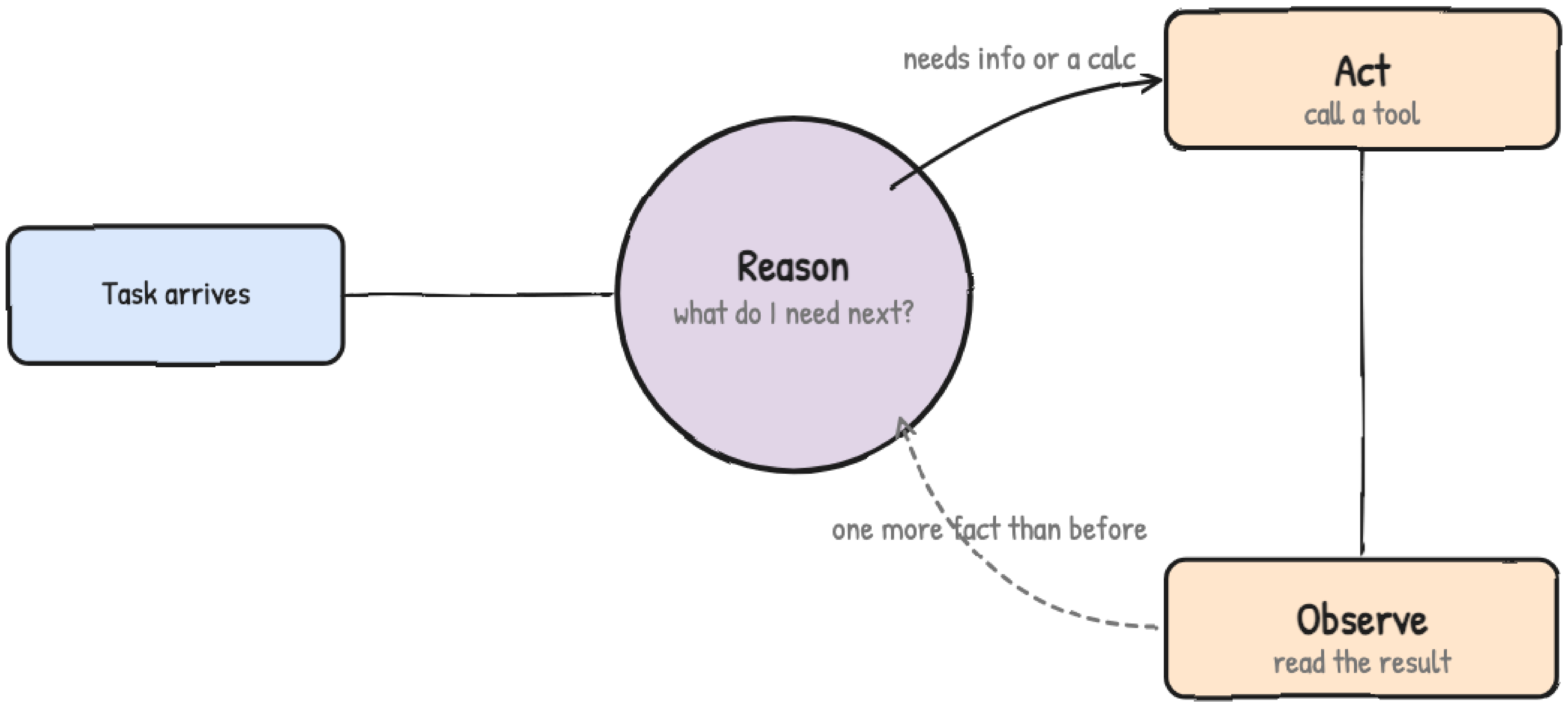

Every real agent is running this underneath

# One loop, run again and again, until the model has enough



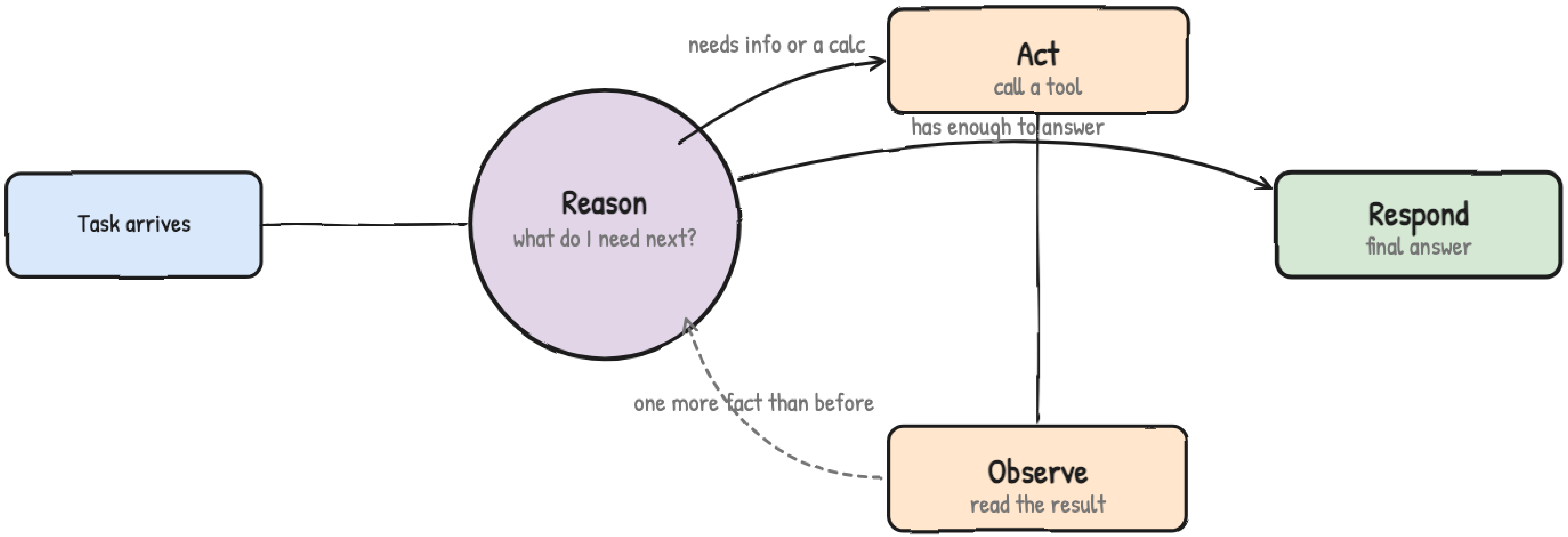

Every real agent is running this underneath

# One loop, run again and again, until the model has enough



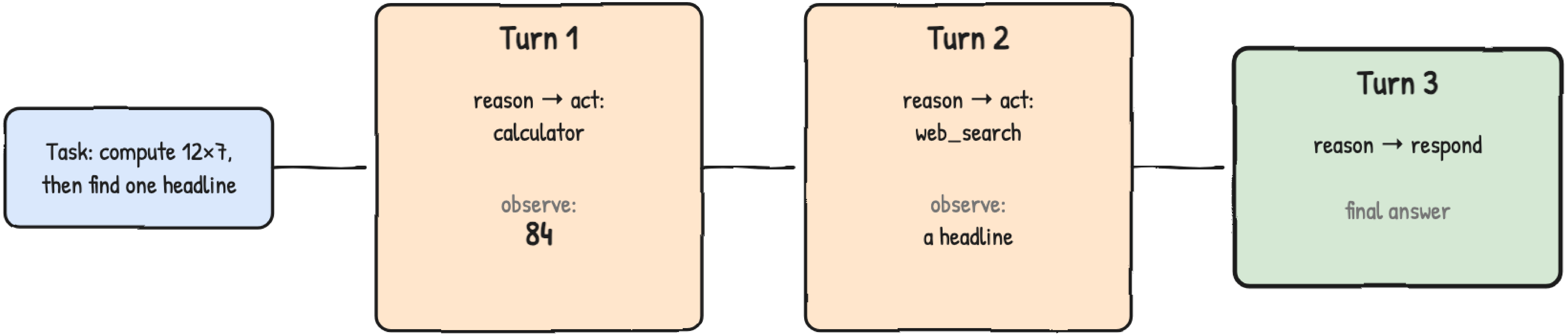

Every real agent is running this underneath

# One loop, run again and again, until the model has enough




Every real agent is running this underneath

# Watch two real tool calls turn into one answer




The exact shape of Lab 4's research agent

# Tools and function calling: how a model decides to act

# A contractor doesn't do the plumbing — they read the note and call

One short note tells you who to call

# A tool schema is a name, a plain description, and the shape of its arguments

handed to the model alongside the conversation

**calculator**

name: "calculator"

description: evaluates a
math expression

arguments: expression (text)

**web_search**

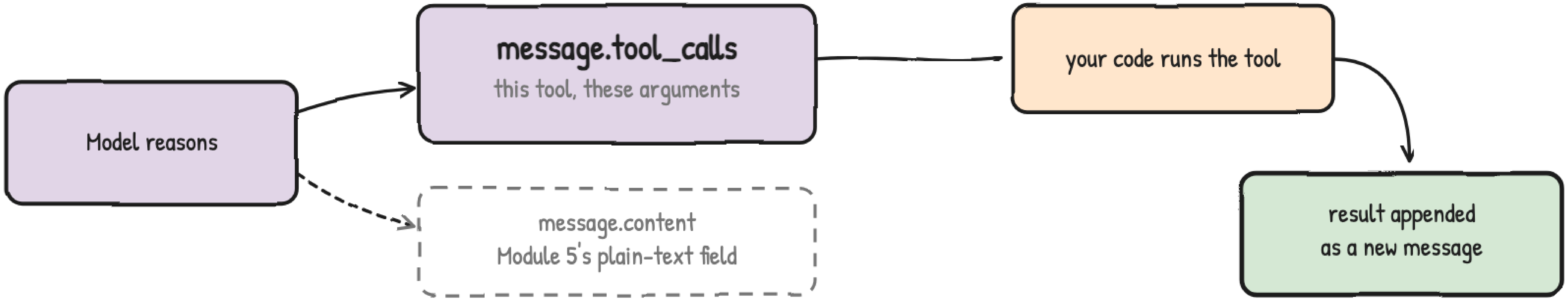
name: "web_search"

description: searches the
web for current information

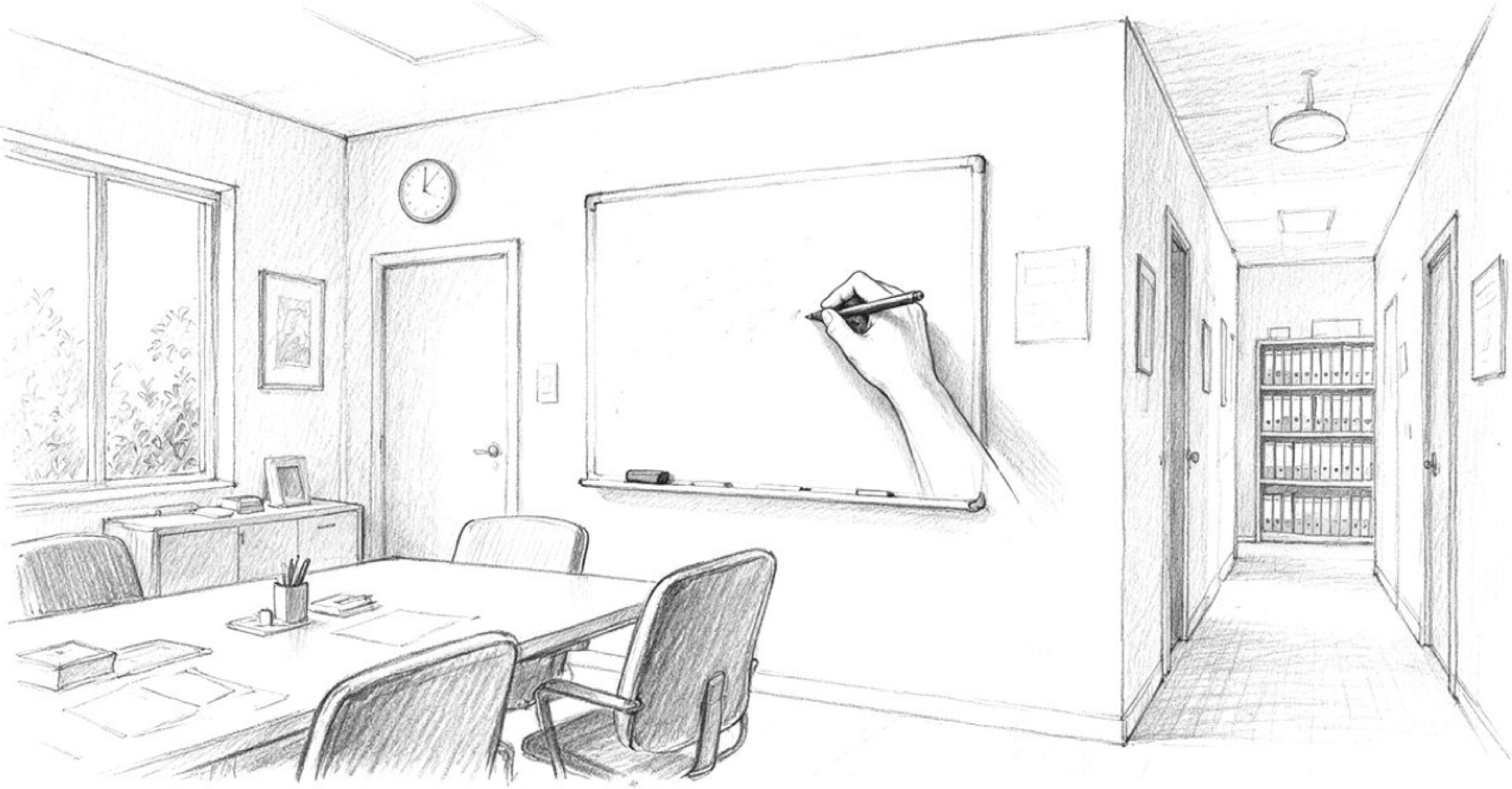
arguments: query, max_results

calculator and web_search — the lab's two tools

# The model writes a request; your code is the one that actually acts

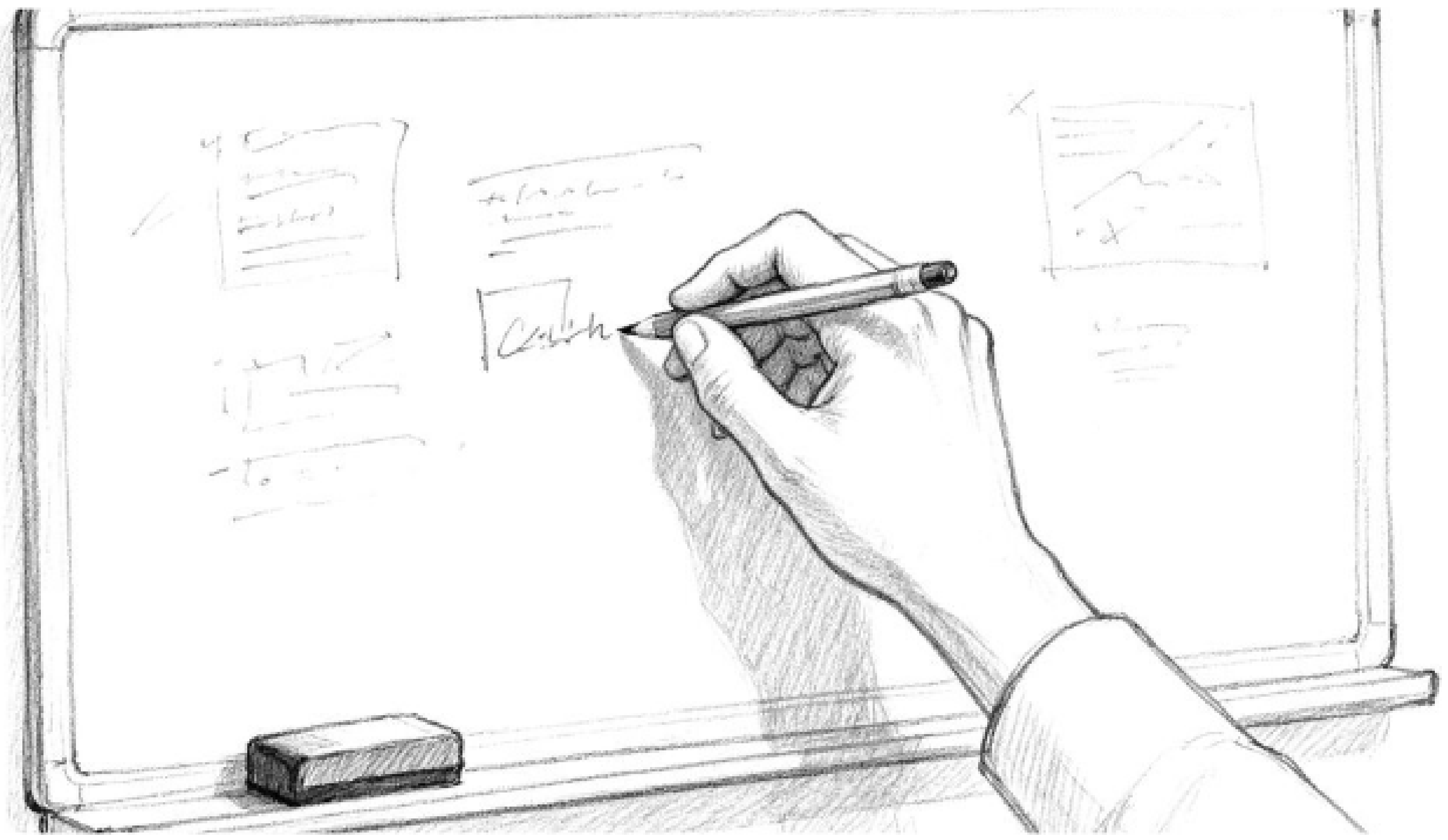



Same response object as Module 5, one new field to check

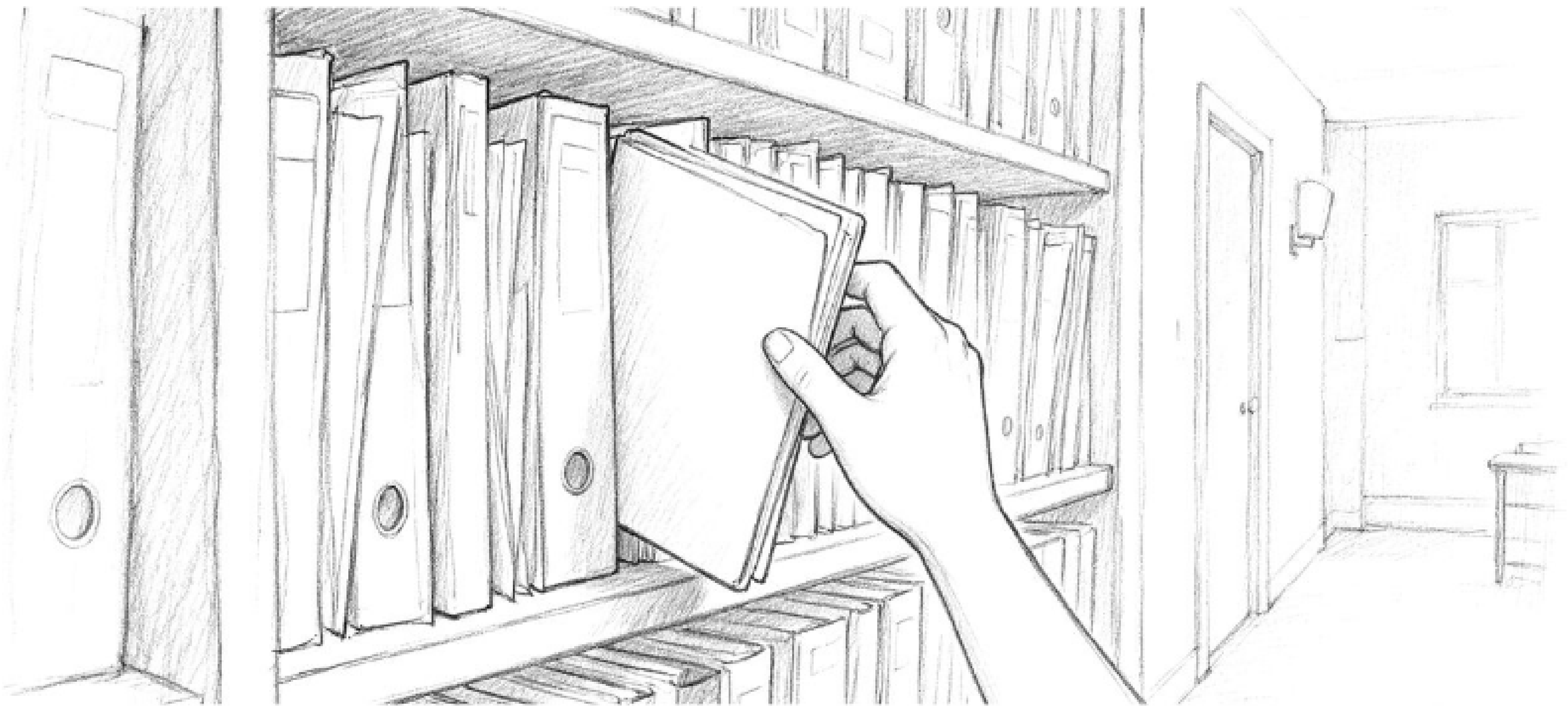


# Memory and planning: context window vs. vector DB

# The whiteboard in the room, and the shelf down the hall



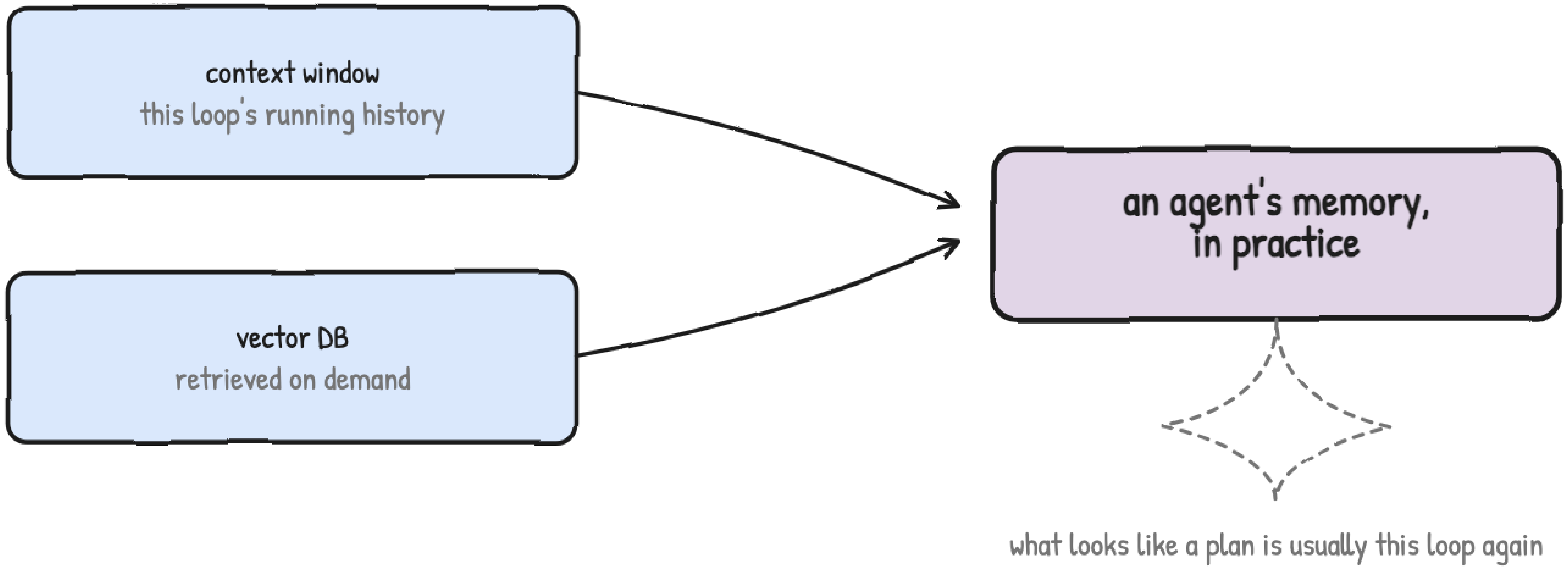
context window — wiped when the exchange ends



vector DB — persists, searched by meaning

Two kinds of memory, side by side

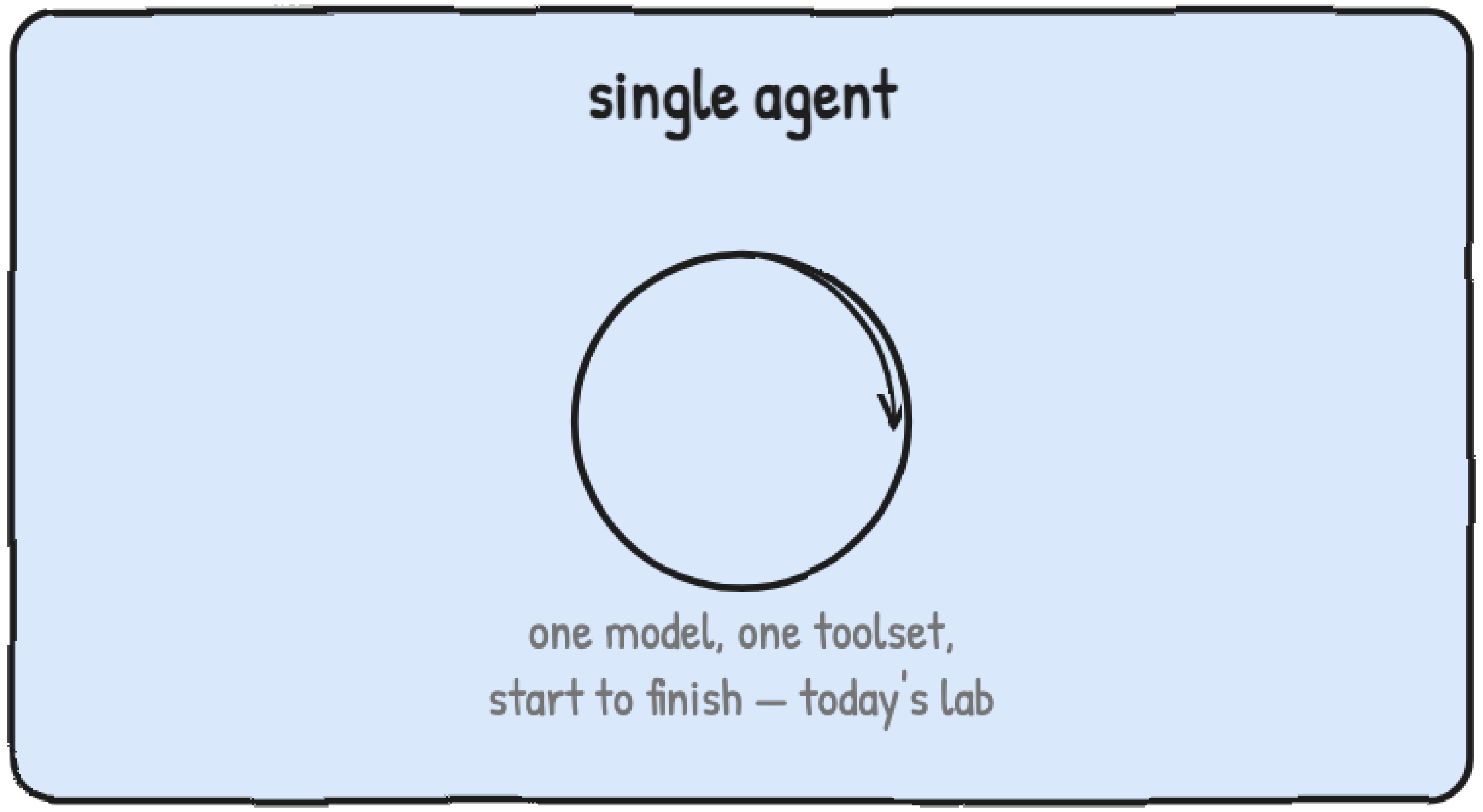
# An agent's memory is almost always a mix of both

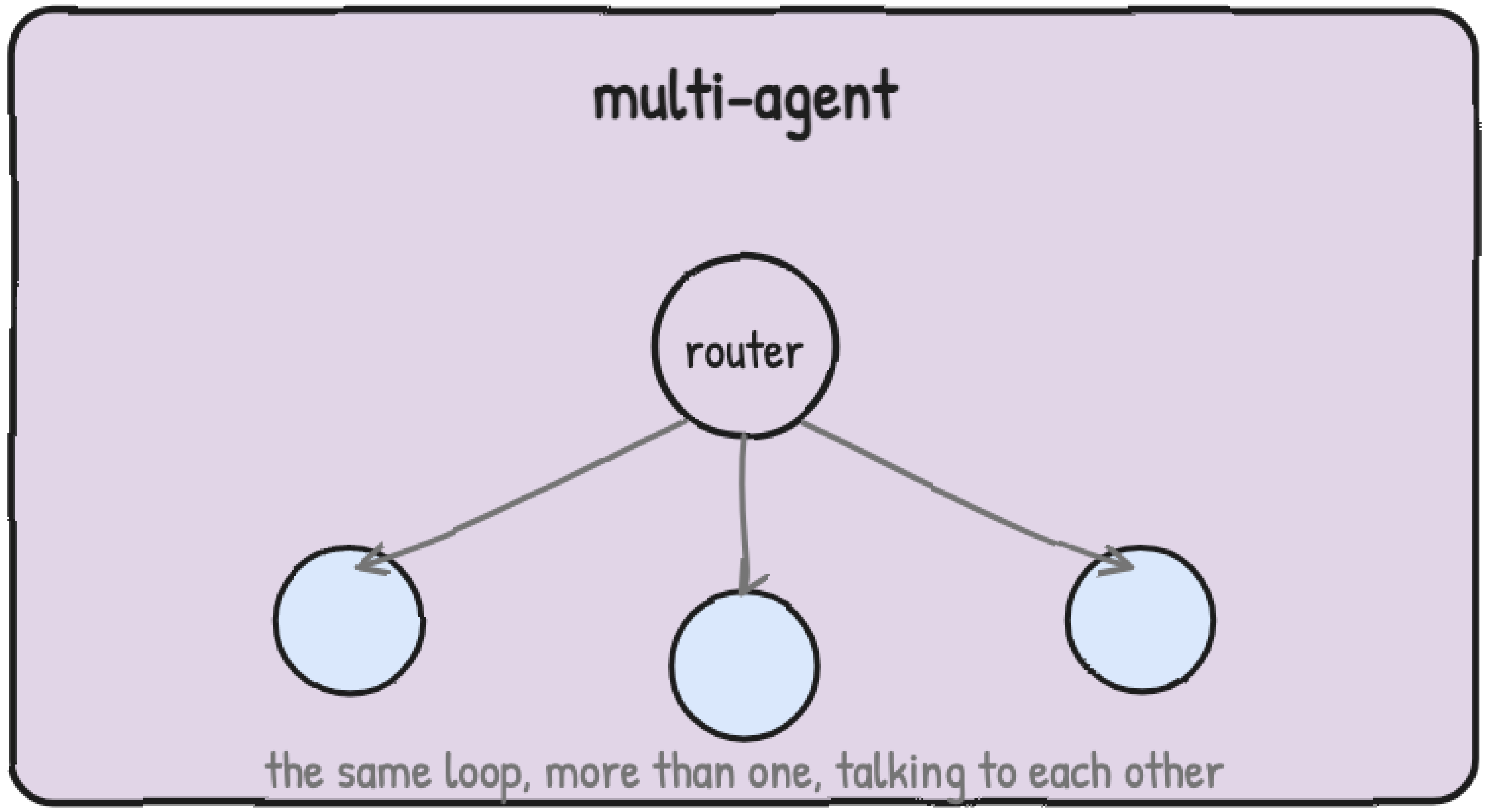



Planning ahead is usually just this loop, one step at a time

Agentic architectures, at a foundational level

# One loop end to end, or a triage desk that routes to specialists




Most of what you'll build is the loop on the left

# Prompt injection: the #1 agent security risk

# A calculator can't hide anything in its answer — a web page can

**calculator**

47 × 12 = 564

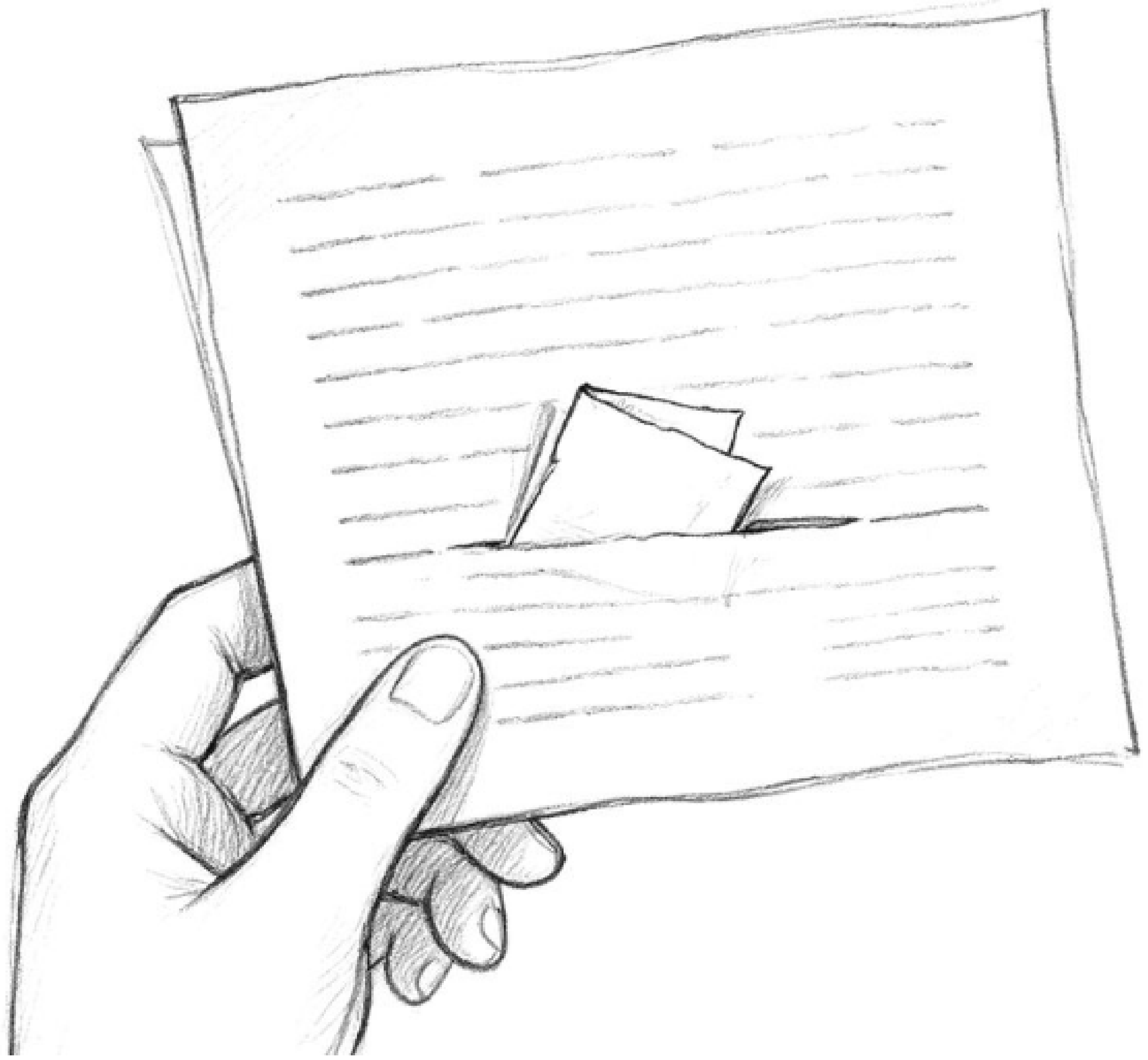
nothing here for anything to hide inside

**web_search**

raw text scraped from the open internet
lands in the conversation as data the model reads next

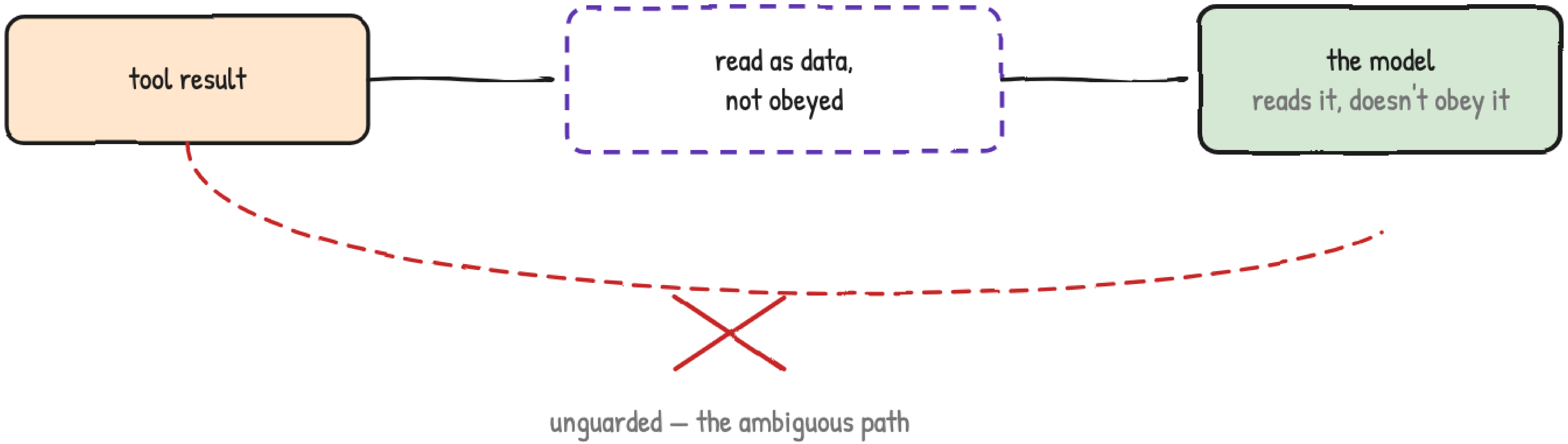
The risk is what the tool returns unfiltered

# Instructions smuggled inside content the agent was only supposed to read



Both arrive as plain text in the same conversation

# Treat everything a tool returns as data, never as an instruction




The deep dive turns this into code, not just a sentence

# How engineers use agents today

# Different tools, same loop

1. agentic coding tools
   → read a codebase, run tests, edit files, across many steps
2. research agents
   → search, cross-reference, summarize — exactly Lab 4
3. operations agents
   → check status, open a ticket, notify the right person

Research agents are what you're about to build

MODULE 6 TAKEAWAYS

# You just watched the loop — now you're about to run it yourself

1. model vs. pipeline vs. agent
2. the agent loop — reason, act, observe, repeat
3. tool schemas and function calling on the wire
4. short-term memory vs. long-term memory
5. prompt injection — the #1 risk, and the mental shift

Now open **Module 6 · Lab** — Build Your First AI Agent. · **Gourav Shah** · School of DevOps & AI